One model, many switches

Measuring the effect of biological realism like for like

Table of contents

Why a second vignette . . . 1
The model . . . 2
State . . . 2
The simplest model . . . 3
The full model . . . 3
Treatment . . . 5
Arrival and fade-out . . . 6
Symbols . . . 6
The outcomes . . . 8
A ladder in one model . . . 8
Every combination . . . 10
The same comparison nearer the threshold . . . 13
A fade-out rule . . . 16
Does the order of adding matter? . . . 19
A stochastic version . . . 20
One run against the equations . . . 21
Fates along the ladder . . . 22
What chance changes . . . 25
Does the size of the population matter? . . . 25
What this shows . . . 27
Limits . . . 27
References . . . 28

Why a second vignette

The first vignette built up realism along a ladder in three host–pathogen systems. Those systems start from three different published models, so their steps cannot be compared directly, and a ladder adds things in one order only.

This vignette uses a single nested model in which each piece of biology is a switch. With every switch off it is the simplest possible model: susceptible and infectious adults. Any combination of switches can be turned on, so that:

- every model is compared with every other like for like;
- the effect of one piece of realism can be measured against every setting of the others;
- the same comparison can be repeated in different conditions.

Switch	What it adds
`latent`	infected animals are not infectious at once
`environment`	half of transmission goes through a pool of zoospores that outlasts the animals that shed them (total transmission unchanged)
`load`	infections start light (little shedding, no disease death) and progress to heavy
`recovery`	infected animals can clear the infection (with `load` on, only light infections)
`larvae`	an aquatic larval stage between birth and adulthood
`larval_infection`	larvae are infected by adults (and from the pool when `environment` is on), without disease, and half carry the infection through metamorphosis

> **!** Important
>
> The model is anchored on the Bsal model of Schmidt et al. (2017) for fire salamanders: with only `latent` on it is their published model, with their parameter values. The carry-over through metamorphosis is from Wang et al. (2021), read from the package's compendium. Everything else is new work in this package and is not a published result, and the values attached to the other switches (pool half-life 30 days, light infections shedding 10% as much and lasting about two weeks (one week when they can also recover), recovery at the same rate as progression, a four-month larval stage) are illustrative.

The model

State

Each animal is an adult or a larva, untreated (u) or treated (t), and in one infection class. Adults are susceptible (S), latent (L), lightly infected (Y) or heavily infected (I). Larvae are uninfected (S^w) or infected (I^w). With the zoospore pool Z this gives 17 variables. Densities are per hectare and time is in years. A class whose switch is off stays empty.

The simplest model

With every switch off and no treatment, there are only susceptible and infectious adults:

$$\frac{dS}{dt} = bS - \beta IS - (d + \gamma N)S,$$
$$\frac{dI}{dt} = \beta IS - d_I I - (d + \gamma N)I,$$

with $N = S + I$ and $\gamma = (b - d)/K$, so that the population without Bsal settles at K. Only susceptible adults reproduce. Turning on `latent` puts the class L between S and I, left at rate e; this is equation 2.1 of Schmidt et al. (2017).

The full model

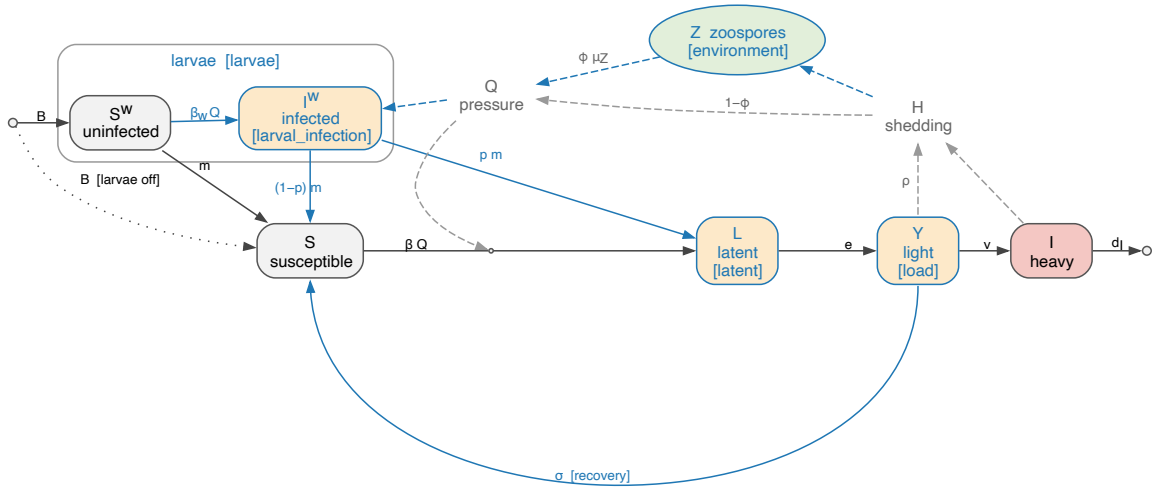


Figure 1: The nested model with every switch on, for untreated animals. Blue boxes and arrows belong to the switch named in brackets; the dotted arrow is used instead of the larval stage when `larvae` is off. Dashed grey arrows are influences, not flows: shedding H by light and heavy adults feeds the infectious pressure Q directly and, with `environment` on, through the zoospore pool Z. Natural deaths (rate D for adults, μw for larvae) leave every box and are not drawn.

Adults shed zoospores, light infections a fraction ρ as much as heavy ones. Adults are infected at rate βQ per susceptible animal, larvae at $\beta_w Q$, where

$$H = \sum_\tau h_\tau \left(\rho Y_\tau + I_\tau\right),$$
$$Q = (1 - \phi)H + \phi \mu_Z Z,$$
$$\frac{dZ}{dt} = H - \mu_Z Z.$$

At equilibrium $\mu_Z Z = H$, so the pool changes the timing of transmission but not its total. Larvae do not shed.

For adults with treatment label $\tau \in \{\mathrm{u}, \mathrm{t}\}$:

$$\begin{aligned}
\frac{dS_\tau}{dt} &= B_\tau^A + M_\tau^S - f_\tau \beta Q S_\tau + \sigma Y_\tau - D S_\tau + \Omega(S_\tau), \\
\frac{dL_\tau}{dt} &= f_\tau \beta Q S_\tau + M_\tau^I - e L_\tau - D L_\tau + \Omega(L_\tau), \\
\frac{dY_\tau}{dt} &= e L_\tau - (\nu + \sigma) Y_\tau - D Y_\tau + \Omega(Y_\tau), \\
\frac{dI_\tau}{dt} &= \nu Y_\tau - k_\tau d_I I_\tau - D I_\tau + \Omega(I_\tau),
\end{aligned}$$

where $D = d + \gamma N_A$ and N_A is the density of all adults. Density dependence acts on adults only. For larvae:

$$\begin{aligned}
\frac{dS_\tau^w}{dt} &= B_\tau^w - f_\tau \beta_w Q S_\tau^w - (m + \mu_w) S_\tau^w + \Omega(S_\tau^w), \\
\frac{dI_\tau^w}{dt} &= f_\tau \beta_w Q S_\tau^w - (m + \mu_w) I_\tau^w + \Omega(I_\tau^w).
\end{aligned}$$

Births are

$$B = b' \sum_\tau \left[S_\tau + Y_\tau + (1 - k_\tau)(L_\tau + I_\tau) \right],$$

and all are untreated: $B_\mathrm{u}^w = B$ with larvae, $B_\mathrm{u}^A = B$ without, and the other birth terms are zero. Without larvae $b' = b$; with them $b' = b(m + \mu_w)/m$, so that each adult still recruits b adults a year when there is no Bsal. Untreated latent and heavily infected adults do not reproduce, as in Schmidt et al. (2017). Light infections do. Treated latent and heavily infected adults reproduce in proportion to the treatment's tolerance.

At metamorphosis (rate m) an infected larva stays infected with probability p and enters the latent class; otherwise it becomes a susceptible adult. Untreated larvae become untreated adults. A fraction r (`retention`) of treated larvae keep the treatment and the rest lose it:

$$\begin{aligned}
M_\mathrm{u}^S &= m \left[S_\mathrm{u}^w + (1 - p) I_\mathrm{u}^w \right] + (1 - r)\, m \left[S_\mathrm{t}^w + (1 - p) I_\mathrm{t}^w \right], \\
M_\mathrm{t}^S &= r\, m \left[S_\mathrm{t}^w + (1 - p) I_\mathrm{t}^w \right], \\
M_\mathrm{u}^I &= p\, m \left[I_\mathrm{u}^w + (1 - r) I_\mathrm{t}^w \right], \\
M_\mathrm{t}^I &= r\, p\, m\, I_\mathrm{t}^w.
\end{aligned}$$

Switching pieces off

Each switch removes terms; nothing else changes.

- `latent` off: L is skipped. New infections, and infected larvae at metamorphosis, enter Y (or I if `load` is also off).
- `load` off: Y is skipped. Latent animals, or new infections, go straight to I, and recovery is from I to S at rate σ. Shedding is $H = \sum_\tau h_\tau I_\tau$. With `load` on, only light infections recover.
- `recovery` off: $\sigma = 0$.
- `environment` off: $\phi = 0$, so $Q = H$ and Z plays no part.
- `larvae` off: there are no larval classes and no M terms; births enter S_u and $b' = b$.
- `larval_infection` off: $\beta_w = 0$, so larvae stay uninfected.

Latent animals and infected larvae never recover, and infected larvae do not die of the disease.

Treatment

Treatment scales three rates. Each effect is multiplied by the protection g that remains:

$$f_\mathrm{u} = h_\mathrm{u} = k_\mathrm{u} = 1, \qquad f_\mathrm{t} = 1 - g\varepsilon, \qquad h_\mathrm{t} = 1 - g\zeta, \qquad k_\mathrm{t} = 1 - g\theta.$$

Efficacy ε cuts the infection rate, tolerance θ cuts disease deaths (the animal stays infectious) and ζ cuts shedding. The block treatment of this vignette is $\varepsilon = 0.99$; the survive treatment is $\theta = 0.99$. Protection can wane with the abundance of the applied bacteria,

$$g(a) = \frac{1 + 1/\kappa}{1 + 2^{a/t_B}/\kappa},$$

where a is the time since the last application, t_B the bacteria's half-life and κ their starting abundance over the abundance that gives half the effect. Without waning, as here, $g = 1$.

Animals also gain and lose the bacteria. For any class X in stage s,

$$\Omega(X_\mathrm{u}) = -\delta F_s X_\mathrm{u} + \omega X_\mathrm{t}, \qquad \Omega(X_\mathrm{t}) = \delta F_s X_\mathrm{u} - \omega X_\mathrm{t},$$

where F_s is the treated fraction of that stage, δ the rate of spread and $\omega = \ln 2/(\text{half-life})$ the rate of loss. Both are zero in this vignette. At each treatment time a fraction c (`coverage`) of the untreated animals of the treated stage becomes treated, and the clock a restarts.

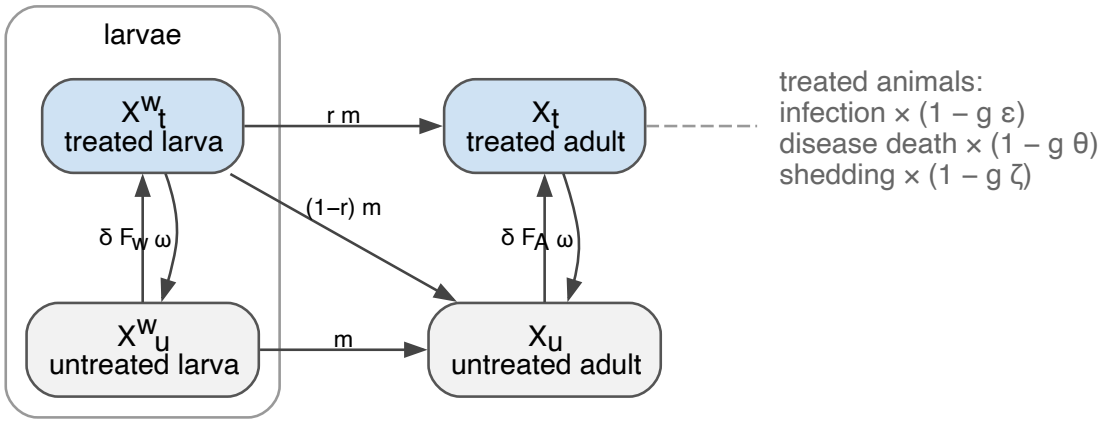


Figure 2: The treatment layer. Every infection class X of the diagram above has an untreated (grey) and a treated (blue) copy. Treated animals gain and lose the bacteria within a stage; at metamorphosis a fraction r of treated larvae stay treated. In this vignette $\delta = \omega = r = 0$ and $g = 1$.

Arrival and fade-out

The population starts without Bsal: $S_u = K$ and, with larvae, $S_u^w = bK/m$. Bsal arrives as $I_0 = 0.01$ heavily infected untreated adults per hectare (one per km², as in Schmidt et al. 2017), after any treatment at the same time. The fade-out rule, when used, is checked after every solver step with threshold $n_{\min}$ (`fadeout_density`): if all animals together fall below $n_{\min}$ they are set to zero; otherwise, once Bsal has arrived, if $0 < \sum(\text{infected animals}) + \mu_Z Z < n_{\min}$, every infected class and the pool are set to zero. The infected animals are removed, not returned to the susceptible class.

The growth rate reported below is the largest real part of the eigenvalues of the Jacobian of the infected classes (L, Y, I, I^w and Z, untreated) at this disease-free state.

Symbols

Symbol	Meaning	Value or source
b	recruitment, per adult	0.39 per yr, compendium (schmidt2017)
d	natural death	0.125 per yr, compendium (schmidt2017)
d_I	Bsal death of heavy infections	52 per yr, compendium (schmidt2017)
e	latent to infectious	364 per yr, compendium (schmidt2017)

Symbol	Meaning	Value or source
β	transmission to adults	52 ha per yr, compendium (schmidt2017)
K	carrying capacity	25 per ha, compendium (schmidt2017); also 3 and 1.5 below
γ	density dependence	$(b - d)/K$
ϕ	share of transmission through the pool	0.5, illustrative
μ_Z	pool decay	ln 2 / 30 days, illustrative
ρ	shedding of a light infection relative to a heavy one	0.1, illustrative
ν	light to heavy	26 per yr, illustrative
σ	recovery	26 per yr, illustrative
m	metamorphosis, 1/(larval period)	1/(0.33 yr), illustrative
μ_w	larval death	1 per yr, illustrative
β_w	transmission from adults to larvae	5.2 ha per yr, illustrative
p	infected larvae still infected after metamorphosis	0.5 (5 of 10), compendium (wang2021)
$\varepsilon, \theta, \zeta$	treatment efficacy, tolerance, shedding reduction	0.99 (block), 0.99 (survive), 0; illustrative
g, κ, t_B	remaining protection and its waning	$g = 1$ (no waning)
δ, ω	spread and loss of the bacteria	0
c, r	coverage, retention at metamorphosis	$c = 1$ when treated, $r = 0$
I_0	arriving infected adults	0.01 per ha
$n_{\min}$	fade-out threshold	10^{-4} per ha, illustrative

```
using Chytridiomycosis, Plots, Printf
default(linewidth = 2, size = (720, 380), legendfontsize = 8, guidefontsize
 ↪  = 9,
    titlefontsize = 10, left_margin = 4Plots.mm, bottom_margin = 4Plots.mm)

T(; kw...) = MicrobiomeTreatment(; kw...)
bio(; kw...) = ChytridBiology(:wang2021; kw...)
schmidt = SchmidtParameters(:schmidt2017)
```

```
SchmidtParameters{Float64}(0.39, 0.125, 52.0, 364.0, 52.0, 25.0, 0.0625, 1.0)
```

The outcomes

Bsal arrives in a population at carrying capacity. Four things are recorded for each model:

- the pathogen's initial growth rate;
- untreated: mean adults over the next ten years, as a fraction of carrying capacity;
- block: the same, with every adult treated once with bacteria that block 99% of infections for life;
- survive: the same, with bacteria that prevent 99% of disease deaths and do not block infection. This is the kind of protection seen in trials.

```
function outcomes(p, b)
    run(tr; kw...) = solve(NestedChytridProblem(p, b, tr; kw...))
    u = run(T())
    bl, sv = run(T(efficacy = 0.99); coverage = 1.0), run(T(tolerance =
 ↪  0.99); coverage = 1.0)
    fate(r) = !isnan(r.host_extinction) ? :host_lost :
     ↪  !isnan(r.pathogen_fadeout) ? :pathogen_gone : :both_persist
    return (growth = invasion_growth_rate(p, b), untreated = u.mean_adults,
     ↪  lowest = u.min_adults,
        block = bl.mean_adults, survive = sv.mean_adults, fates = (fate(u),
 ↪  fate(bl), fate(sv)))
end
```

```
outcomes (generic function with 1 method)
```

A check first: with only the latent class switched on, the growth rate equals the one that follows from the equations of Schmidt et al. (2017).

```
p = schmidt
λ = (sqrt((p.e - p.d_I)^2 + 4 * p.e * p.β * p.K) - 2p.b - p.d_I - p.e) / 2
(nested = invasion_growth_rate(p, bio(latent = true)), schmidt = λ)
```

```
(nested = 496.972318234428, schmidt = 496.97231824502785)
```

A ladder in one model

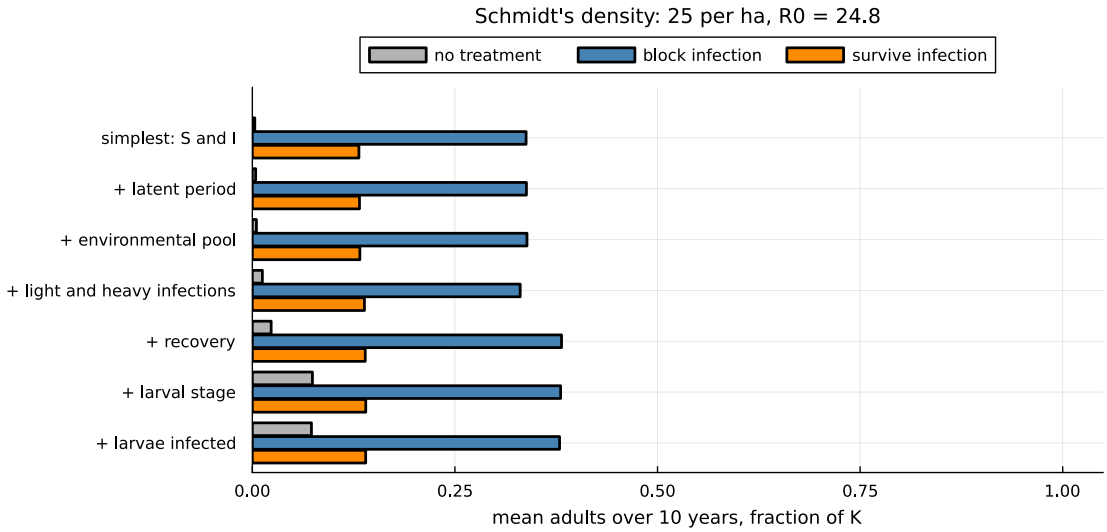
The switches are turned on one after another, starting from the simplest model.

```
order = [:latent, :environment, :load, :recovery, :larvae,
 ↪  :larval_infection]
names_ = ["simplest: S and I", "+ latent period", "+ environmental pool", "+
 ↪  light and heavy infections",
    "+ recovery", "+ larval stage", "+ larvae infected"]
ladder_models(p) = [outcomes(p, bio(; (k => true for k in order[1:i])...))
 ↪  for i in 0:length(order)]

function ladder_figure(rows, title)
    k = length(rows)
    bar((1:k) .- 0.27, [r.untreated for r in rows]; orientation = :h,
     ↪  bar_width = 0.25, label = "no treatment",
        color = :gray70, yticks = (1:k, names_), yflip = true, xlims = (0,
 ↪  1.05), xticks = 0:0.25:1,
        xlabel = "mean adults over 10 years, fraction of K", size = (840, 80
 ↪  + 44k),
        left_margin = 6Plots.mm, legend = :outertop, legend_column = 3,
 ↪  title)
    bar!(1:k, [r.block for r in rows]; orientation = :h, bar_width = 0.25,
     ↪  label = "block infection", color = :steelblue)
    bar!((1:k) .+ 0.27, [r.survive for r in rows]; orientation = :h,
     ↪  bar_width = 0.25, label = "survive infection",
        color = :darkorange)
end

function ladder_table(rows)
    println("| Model | Growth rate (per yr) | Untreated | Lowest untreated |
     ↪  Block | Survive |")
    println("|---|---|---|---|---|---|")
    for (n, r) in zip(names_, rows)
        @printf("| %s | %.0f | %.3f | %.0e | %.3f | %.3f |\n", n, r.growth,
         ↪  r.untreated, r.lowest, r.block, r.survive)
    end
end

high = ladder_models(schmidt)
ladder_figure(high, "Schmidt's density: 25 per ha, R0 = 24.8")
```




```
ladder_table(high)
```

Model	Growth rate (per yr)	Untreated	Lowest untreated	Block	Survive
simplest: S and I	1248	0.003	2e-11	0.338	0.132
+ latent period	497	0.004	2e-11	0.338	0.132
+ environmental pool	309	0.005	2e-11	0.339	0.133
+ light and heavy infections	111	0.012	6e-06	0.331	0.138
+ recovery	98	0.023	2e-03	0.382	0.139
+ larval stage	98	0.074	2e-02	0.380	0.140
+ larvae infected	98	0.073	2e-02	0.379	0.140

At the density Schmidt et al. used, where Bsal's reproduction number is about 25, the added biology changes the conclusions very little. The epidemic slows more than ten-fold as switches are turned on, but the population collapses in every model, and each treatment gives almost the same result throughout. One thing does change: the lowest density the untreated population reaches rises by many orders of magnitude, which matters for whether it could go extinct.

Every combination

A ladder is one path. The four switches for the adult stage are now set in every combination, with three settings for the larvae (none, present but not infected, infected): 48 models.

```
configs = vec([(; latent, environment, load, recovery, larvae = lv > 0,
 ↪  larval_infection = lv > 1)
    for latent in (false, true), environment in (false, true), load in
 ↪  (false, true),
        recovery in (false, true), lv in 0:2])
all_models(p; fade = 0.0) = [(c, outcomes(p, bio(; fadeout_density = fade,
 ↪  c...))) for c in configs]

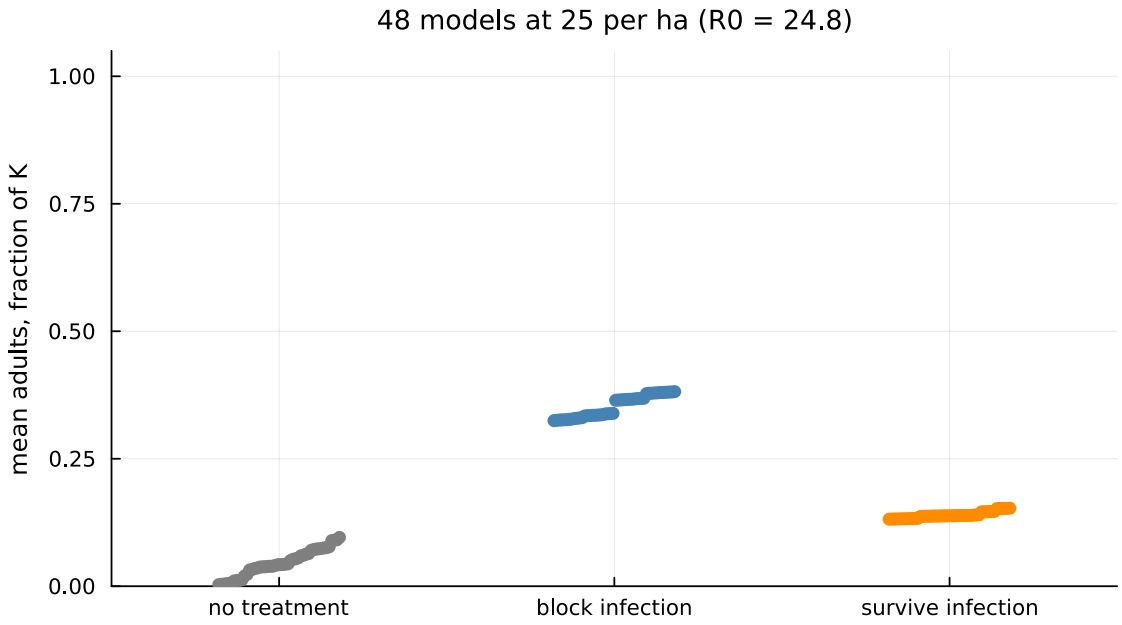
same(c, c2, skip) = all(getfield(c, k) == getfield(c2, k) for k in keys(c)
 ↪  if !(k in skip))
pieces = [
    ("latent period", c -> c.latent, (c, c2) -> !c2.latent && same(c, c2,
 ↪  (:latent,))),
    ("environmental pool", c -> c.environment, (c, c2) -> !c2.environment &&
 ↪  same(c, c2, (:environment,))),
    ("light and heavy infections", c -> c.load, (c, c2) -> !c2.load &&
 ↪  same(c, c2, (:load,))),
    ("recovery", c -> c.recovery, (c, c2) -> !c2.recovery && same(c, c2,
 ↪  (:recovery,))),
    ("larval stage", c -> c.larvae && !c.larval_infection,
        (c, c2) -> !c2.larvae && same(c, c2, (:larvae, :larval_infection))),
    ("larvae infected", c -> c.larval_infection,
        (c, c2) -> c2.larvae && !c2.larval_infection && same(c, c2,
 ↪  (:larvae, :larval_infection))),
]
# change in an outcome when a piece is switched on, for every setting of the
 ↪  other pieces
effects(res, on, off, f) = [f(o) - f(o2) for (c, o) in res if on(c) for (c2,
 ↪  o2) in res if off(c, c2)]

function effects_table(res)
    println("| Piece switched on | Untreated: mean (range) | Block: mean
     ↪  (range) | Survive: mean (range) |")
    println("|---|---|---|---|")
    for (label, on, off) in pieces
        cells = map((o -> o.untreated, o -> o.block, o -> o.survive)) do f
            d = effects(res, on, off, f)
            @sprintf("%+.2f (%+.2f to %+.2f)", sum(d) / length(d),
             ↪  minimum(d), maximum(d))
        end
        println("| ", label, " | ", join(cells, " | "), " |")
    end
end

function spread_figure(res, title)
```

```
    f = (o -> o.untreated, o -> o.block, o -> o.survive)
    labels = ["no treatment", "block infection", "survive infection"]
    plot(; xticks = (1:3, labels), xlims = (0.5, 3.5), ylims = (0, 1.05),
     ↪  ylabel = "mean adults, fraction of K", legend = false,
        title, size = (640, 360))
    for (i, g) in enumerate(f)
        v = [g(o) for (_, o) in res]
        scatter!(i .+ range(-0.18, 0.18; length = length(v)), sort(v);
         ↪  markersize = 4,
            color = (:gray50, :steelblue, :darkorange)[i], markerstrokewidth
 ↪  = 0)
    end
    current()
end

res_high = all_models(schmidt)
spread_figure(res_high, "48 models at 25 per ha (R0 = 24.8)")
```




```
effects_table(res_high)
```

Piece switched on	Untreated: mean (range)	Block: mean (range)	Survive: mean (range)
latent period	+0.00 (+0.00 to +0.00)	+0.00 (+0.00 to +0.00)	+0.00 (+0.00 to +0.01)

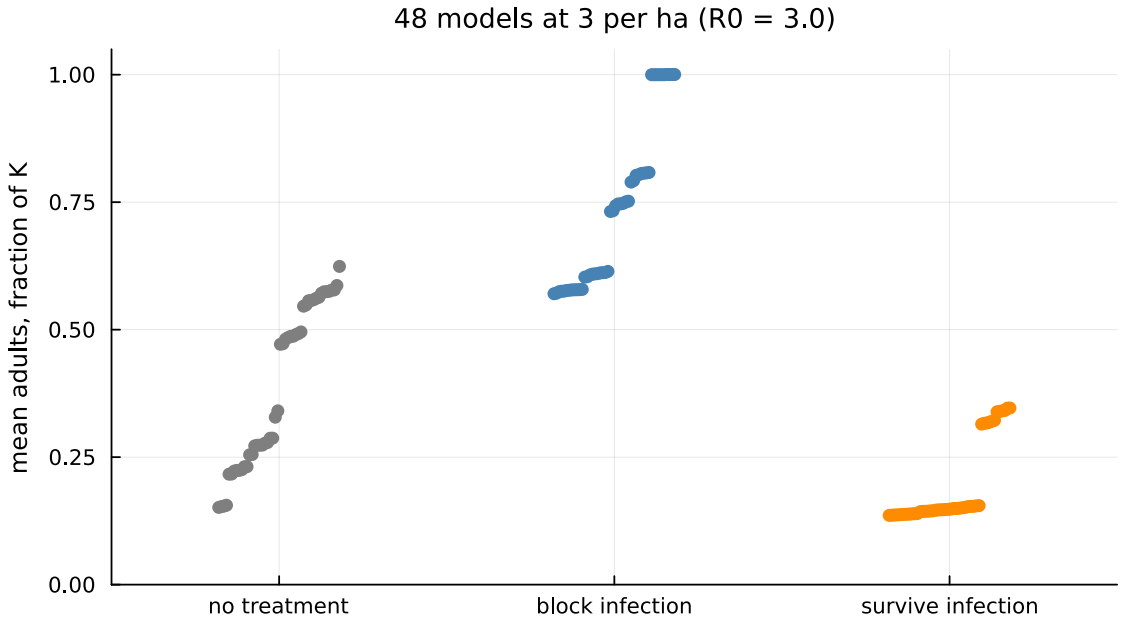
Piece switched on	Untreated: mean (range)	Block: mean (range)	Survive: mean (range)
environmental pool	-0.02 (-0.08 to +0.00)	+0.00 (+0.00 to +0.00)	+0.00 (-0.00 to +0.00)
light and heavy infections	+0.01 (-0.00 to +0.02)	+0.00 (-0.01 to +0.01)	-0.00 (-0.01 to +0.01)
recovery	+0.03 (+0.00 to +0.09)	+0.04 (+0.03 to +0.05)	+0.01 (+0.00 to +0.02)
larval stage	+0.02 (-0.03 to +0.05)	-0.00 (-0.00 to -0.00)	+0.00 (+0.00 to +0.00)
larvae infected	-0.00 (-0.01 to -0.00)	-0.00 (-0.00 to -0.00)	-0.00 (-0.00 to -0.00)

Each dot is one of the 48 models. At this density they sit almost on top of each other: no piece of host biology moves any outcome by more than about 0.1. When the pathogen spreads this fast and kills this quickly, the simple model gives the same answer as the complex one.

The same comparison nearer the threshold

Schmidt et al. show that Bsal cannot invade below about one adult female per hectare. The comparison is repeated at 3 per hectare (reproduction number 3) and 1.5 per hectare (1.5).

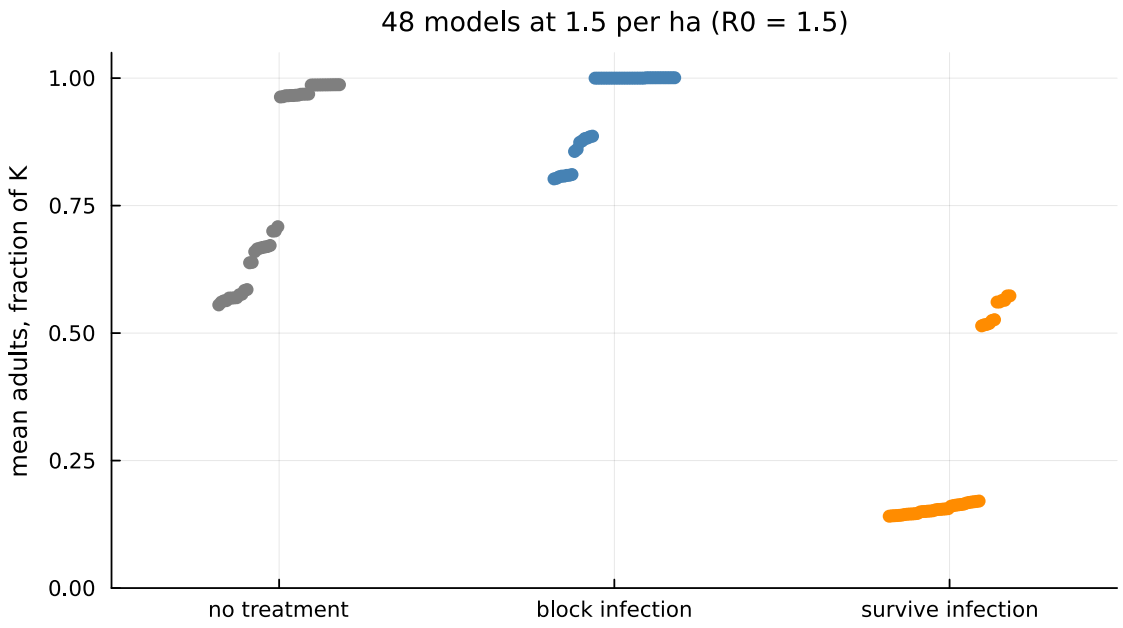
```
mid, low = Chytridiomycosis.with(schmidt; K = 3.0),
 ↪  Chytridiomycosis.with(schmidt; K = 1.5)
res_mid, res_low = all_models(mid), all_models(low)
spread_figure(res_mid, "48 models at 3 per ha (R0 = 3.0)")
```



```
effects_table(res_mid)
```

Piece switched on	Untreated: mean (range)	Block: mean (range)	Survive: mean (range)
latent period	+0.01 (-0.00 to +0.09)	+0.03 (-0.00 to +0.39)	+0.00 (+0.00 to +0.00)
environmental pool	-0.02 (-0.09 to +0.03)	-0.10 (-0.39 to +0.01)	-0.00 (-0.03 to +0.00)
light and heavy infections	+0.01 (-0.07 to +0.09)	-0.03 (-0.42 to +0.06)	-0.09 (-0.20 to +0.01)
recovery	+0.30 (+0.17 to +0.47)	+0.22 (+0.00 to +0.42)	+0.10 (+0.01 to +0.21)
larval stage	+0.04 (-0.06 to +0.13)	+0.00 (-0.00 to +0.00)	-0.00 (-0.01 to +0.00)
larvae infected	-0.03 (-0.11 to -0.00)	-0.09 (-0.40 to -0.00)	-0.00 (-0.00 to -0.00)

```
spread_figure(res_low, "48 models at 1.5 per ha (R0 = 1.5)")
```



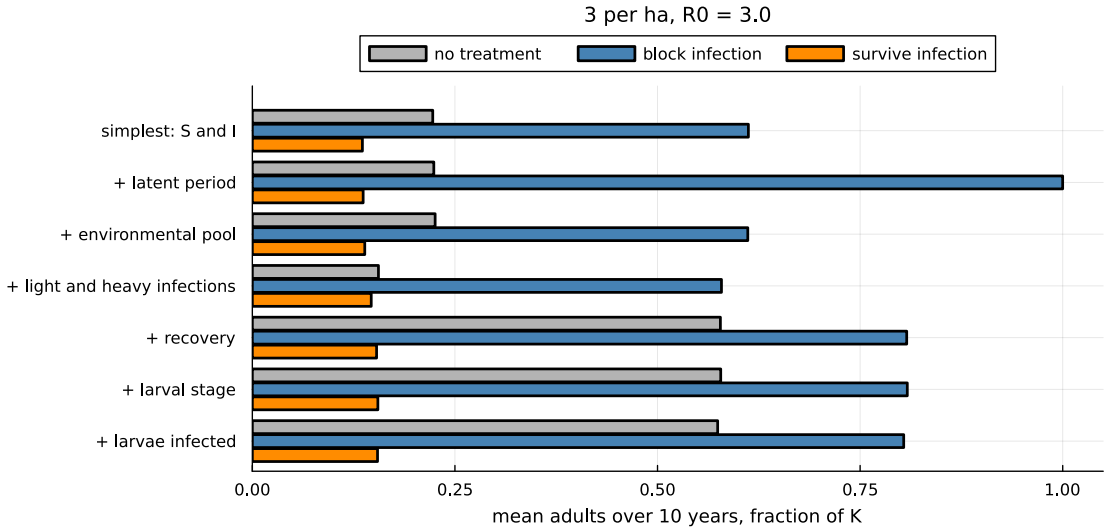

```
effects_table(res_low)
```

Piece switched on	Untreated: mean (range)	Block: mean (range)	Survive: mean (range)
latent period	+0.00 (+0.00 to +0.05)	+0.00 (-0.00 to +0.00)	+0.00 (-0.00 to +0.00)
environmental pool	-0.00 (-0.04 to +0.03)	-0.05 (-0.19 to +0.02)	-0.01 (-0.05 to +0.01)
light and heavy infections	-0.04 (-0.12 to +0.02)	-0.02 (-0.08 to -0.00)	-0.18 (-0.41 to +0.01)
recovery	+0.36 (+0.26 to +0.43)	+0.11 (+0.00 to +0.20)	+0.21 (+0.01 to +0.43)
larval stage	-0.00 (-0.04 to +0.01)	+0.00 (-0.00 to +0.00)	-0.00 (-0.01 to +0.00)
larvae infected	-0.01 (-0.07 to -0.00)	-0.04 (-0.19 to -0.00)	-0.00 (-0.00 to -0.00)

Now the models disagree. At 3 per hectare the untreated outcome ranges from 0.15 to 0.62 depending on which biology is included, and the infection-blocking treatment from 0.57 to full protection. Realism matters most near the threshold.

The ladder at 3 per hectare:

```
ladder_figure(ladder_models(mid), "3 per ha, R0 = 3.0")
```



Three things stand out.

1. Recovery is the piece that matters most, for the untreated population and for the blocking treatment. This project has no estimate of it for Bsal.
2. Pieces do not have a fixed effect. The ranges in the tables are wide: an environmental pool, or infected larvae, changes the blocking treatment by anything from nothing to −0.4 depending on what else is in the model.

3. Bacteria that let animals survive infection do worse than no treatment at these densities, in every one of the 48 models (orange below grey; counts in the table below). Untreated, an epidemic at low density burns out and the population recovers. Treated animals that survive stay infectious in this model, and keep infecting the untreated animals born after them.

```
println("| Density | Models in which the survive treatment does worse than
 ↪  none | Blocking does worse than none |")
println("|---|---|---|")
for (label, res) in (("25 per ha", res_high), ("3 per ha", res_mid), ("1.5
 ↪  per ha", res_low))
    @printf("| %s | %d of 48 | %d of 48 |\n", label, count(x -> x[2].survive
     ↪  < x[2].untreated, res),
        count(x -> x[2].block < x[2].untreated - 1e-6, res))
end
```

Density	Models in which the survive treatment does worse than none	Blocking does worse than none
25 per ha	0 of 48	0 of 48
3 per ha	48 of 48	0 of 48
1.5 per ha	48 of 48	0 of 48

A fade-out rule

The equations are deterministic, so a density can fall to a value that means far less than one animal and then grow back. Near the threshold that can decide the result. The model is therefore given a fade-out rule, itself a piece of realism: once infected animals (adults and larvae, with the zoospore pool counted in infectious-animal equivalents) fall below one per 100 km² (0.0001 per hectare) the pathogen is removed, and if all animals fall below that density the host population is removed. The threshold is illustrative.

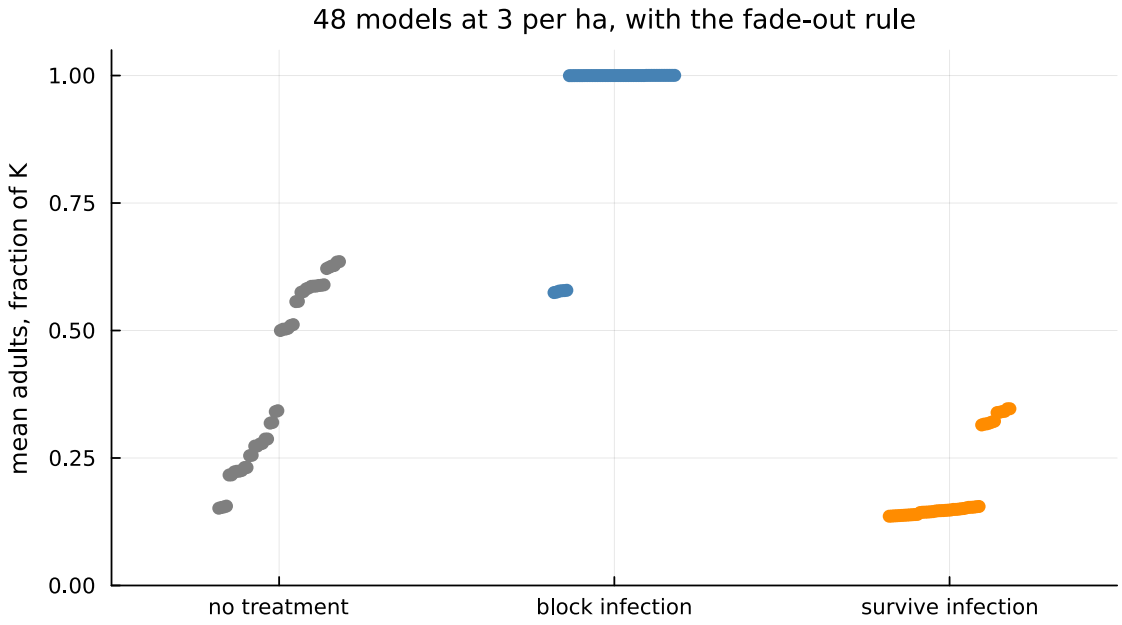
```
fade = 1e-4
fres_high, fres_mid, fres_low = all_models(schmidt; fade), all_models(mid;
 ↪  fade), all_models(low; fade)
nothing
```

With the rule, each run has a fate within the ten years: the host is lost, the pathogen is lost and the host remains, or both persist.

```
println("| Density | Treatment | Host lost | Pathogen lost, host remains |
 ↪  Both persist | Mean adults: range |")
println("|---|---|---|---|---|---|")
for (label, res) in (("25 per ha", fres_high), ("3 per ha", fres_mid), ("1.5
 ↪  per ha", fres_low))
    for (i, (name, f)) in enumerate((("none", o -> o.untreated), ("block", o
     ↪  -> o.block), ("survive", o -> o.survive)))
        fates = [o.fates[i] for (_, o) in res]
        v = [f(o) for (_, o) in res]
        @printf("| %s | %s | %d | %d | %d | %.2f to %.2f |\n", label, name,
         ↪  count(==(:host_lost), fates),
            count(==(:pathogen_gone), fates), count(==(:both_persist),
             ↪  fates), minimum(v), maximum(v))
    end
end
```

Density	Treatment	Host lost	Pathogen lost, host remains	Both persist	Mean adults: range
25 per ha	none	4	34	10	0.00 to 0.10
25 per ha	block	0	0	48	0.32 to 0.38
25 per ha	survive	0	0	48	0.13 to 0.15
3 per ha	none	0	48	0	0.15 to 0.64
3 per ha	block	0	42	6	0.57 to 1.00
3 per ha	survive	0	12	36	0.14 to 0.35
1.5 per ha	none	0	48	0	0.56 to 0.99
1.5 per ha	block	0	48	0	1.00 to 1.00
1.5 per ha	survive	0	12	36	0.14 to 0.57

```
spread_figure(fres_mid, "48 models at 3 per ha, with the fade-out rule")
```




```
effects_table(fres_mid)
```

Piece switched on	Untreated: mean (range)	Block: mean (range)	Survive: mean (range)
latent period	+0.00 (-0.00 to +0.00)	+0.00 (-0.00 to +0.00)	+0.00 (+0.00 to +0.00)
environmental pool	-0.04 (-0.08 to +0.00)	-0.11 (-0.43 to -0.00)	-0.00 (-0.03 to +0.00)
light and heavy infections	+0.00 (-0.07 to +0.09)	-0.11 (-0.43 to -0.00)	-0.09 (-0.20 to +0.01)
recovery	+0.33 (+0.22 to +0.47)	+0.11 (+0.00 to +0.43)	+0.10 (+0.01 to +0.21)
larval stage	+0.05 (-0.00 to +0.13)	-0.00 (-0.00 to +0.00)	-0.00 (-0.01 to +0.00)
larvae infected	-0.01 (-0.02 to -0.01)	-0.00 (-0.00 to -0.00)	-0.00 (-0.00 to -0.00)

What the rule changes, and what it does not:

1. One apparent effect was an artefact. Without the rule, adding a latent period at 3 per hectare seemed to turn the blocking treatment from partial to full protection (+0.39). With the rule the latent period has no effect at all. In both models the pathogen is in fact lost; the model without the latent period was regrowing it from a meaningless density.
2. The blocking treatment now eliminates the pathogen in most models near the threshold: in 42 of 48 at 3 per hectare and in all 48 at 1.5. The exceptions at 3 per hectare are models with an environmental pool and light infections but no recovery.
3. Recovery remains the piece that matters most for the untreated population.

4. Treatment changes the fate as well as the numbers. At 25 per hectare an untreated epidemic burns itself out in most models, and in a few the host is lost. With either treatment, host and pathogen both persist in every model: the treated animals keep the population going, and with it the pathogen.
5. The survive-type treatment still does worse than none at the lower densities. In most models it is the reason the pathogen persists.

Does the order of adding matter?

The effect of each piece when it is the first thing added to the simplest model, and when it is the last thing added to the otherwise complete model, at 3 per hectare with the fade-out rule:

```
full = (latent = true, environment = true, load = true, recovery = true,
 ↪  larvae = true, larval_infection = true)
empty_ = (latent = false, environment = false, load = false, recovery =
 ↪  false, larvae = false, larval_infection = false)
get_(res, c) = only(o for (c2, o) in res if c2 == c)
println("| Piece | Untreated: added first | added last | Block: added first
 ↪  | added last |")
println("|---|---|---|---|---|")
for (label, on, off) in (("latent period", (latent = true,), (latent =
 ↪  false,)),
        ("environmental pool", (environment = true,), (environment =
 ↪  false,)),
        ("light and heavy infections", (load = true,), (load = false,)),
        ("recovery", (recovery = true,), (recovery = false,)),
        ("larval stage, infected", (larvae = true, larval_infection = true),
 ↪  (larvae = false, larval_infection = false)))
    first_ = (get_(fres_mid, merge(empty_, on)), get_(fres_mid, empty_))
    last_ = (get_(fres_mid, full), get_(fres_mid, merge(full, off)))
    @printf("| %s | %+.2f | %+.2f | %+.2f | %+.2f |\n", label,
     ↪  first_[1].untreated - first_[2].untreated,
        last_[1].untreated - last_[2].untreated, first_[1].block -
 ↪  first_[2].block, last_[1].block - last_[2].block)
end
```

Piece	Untreated: added first	added last	Block: added first	added last
latent period	+0.00	+0.00	-0.00	-0.00
environmental pool	+0.00	-0.04	-0.00	-0.00
light and heavy infections	-0.07	+0.08	-0.00	-0.00

Piece	Untreated: added first	added last	Block: added first	added last
recovery	+0.33	+0.37	+0.00	+0.42
larval stage, infected	+0.10	-0.01	-0.00	-0.00

Where the two columns differ, the piece interacts with the others, and a ladder would give a different impression depending on where the piece was placed.

A stochastic version

A fade-out rule is a crude stand-in for chance. The model also exists in a stochastic form, `StochasticNestedProblem`, in which animals are whole numbers in a given area and every transition happens at random. It draws its events from the same list of transitions as the equations, so with a large enough area it reproduces them, and with one infected animal its chance of a major outbreak matches the theoretical $1 - 1/R_0$ (both are checked in the package tests). Bsal now arrives as one infected adult in 20 hectares: 500 adults at Schmidt's density, 60 at 3 per hectare and 30 at 1.5. Each model is run 100 times.

The stochastic version counts animals, $n_i = A x_i$ in an area A. Time moves in steps of $\Delta t = 6$ hours. At the start of each step every flow of the equations above is turned into a rate per animal, λ_{ij} from class i to destination j (a death is one destination). Then

$$O_i \sim \text{Binomial}\left(n_i,\ 1 - e^{-\Lambda_i \Delta t}\right), \qquad \Lambda_i = \sum_j \lambda_{ij},$$

$$(O_{i\to j})_j \sim \text{Multinomial}\left(O_i,\ (\lambda_{ij}/\Lambda_i)_j\right),$$

$$\text{births} \sim \text{Poisson}\left(AB\Delta t\right).$$

The pool stays continuous and is updated exactly with shedding held at its value at the start of the step: $Z \leftarrow Ze^{-\mu_Z \Delta t} + (H/\mu_Z)(1 - e^{-\mu_Z \Delta t})$. At a treatment each untreated animal is treated with probability c. The fade-out rule is not used. Bsal is lost when no animal is infected and the pool is expected to cause fewer than 0.05 more infections ($\beta\phi ZS < 0.05$, with S the number of susceptible animals); the pool is then emptied.

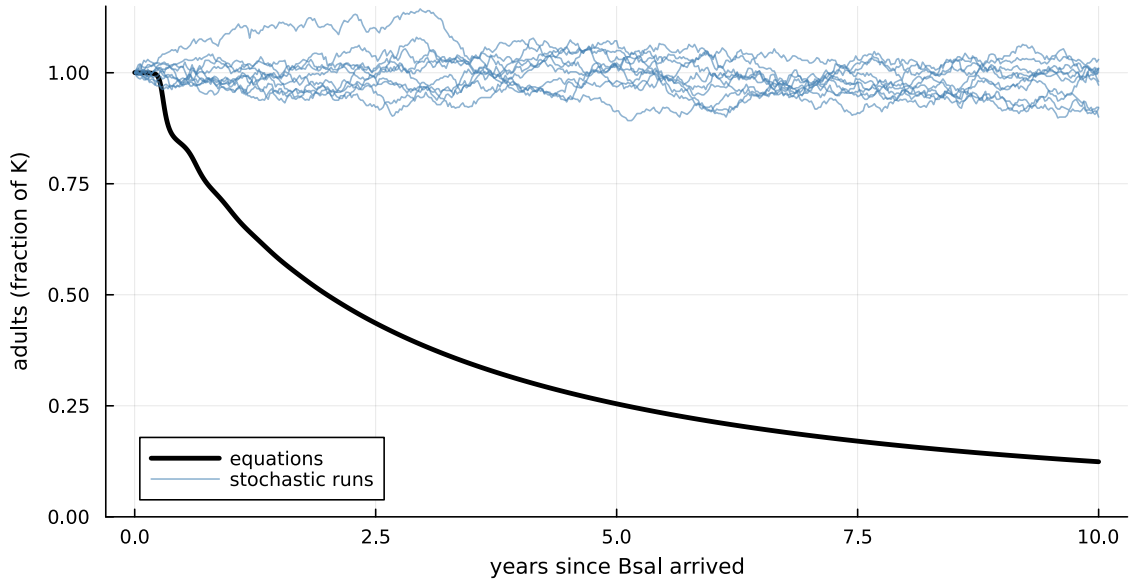
```
using Random: Xoshiro
reps, area = 100, 20.0
function stochastic_fates(p, b, tr, cov; area = area)
    runs = [solve(StochasticNestedProblem(p, b, tr; area, coverage = cov);
 ↪  rng = Xoshiro(s)) for s in 1:reps]
    host = count(r -> !isnan(r.host_extinction), runs) / reps
    gone = count(r -> isnan(r.host_extinction) &&
 ↪  !isnan(r.pathogen_extinction), runs) / reps
    return (host = host, gone = gone, both = 1 - host - gone,
```

```
        minor = count(r -> r.peak_infected < 10, runs) / reps,
        mean = sum(r.mean_adults for r in runs) / reps)
end
ladder_bios = [bio(; (k => true for k in order[1:i])...) for i in
 ↪  0:length(order)]
treatments = (("no treatment", T(), 0.0), ("block infection", T(efficacy =
 ↪  0.99), 1.0),
    ("survive infection", T(tolerance = 0.99), 1.0))
densities = (("25 per ha", schmidt), ("3 per ha", mid), ("1.5 per ha", low))
sto = Dict((d, t, i) => stochastic_fates(p, b, tr, cov)
    for (d, p) in densities for (t, tr, cov) in treatments for (i, b) in
     ↪  enumerate(ladder_bios))
nothing
```

One run against the equations

At Schmidt's density, with every adult treated on arrival with bacteria that block 99% of infections, the equations give about a third of the population after ten years (step 1 of the ladder above). Ten stochastic runs of the same model:

```
b1 = bio(latent = true)
det = solve(NestedChytridProblem(schmidt, b1, T(efficacy = 0.99); coverage =
 ↪  1.0))
plt = plot(det.t, det.adults ./ schmidt.K; color = :black, linewidth = 3,
 ↪  label = "equations",
    xlabel = "years since Bsal arrived", ylabel = "adults (fraction of K)",
 ↪  ylims = (0, 1.15),
    legend = :bottomleft)
for s in 1:10
    r = solve(StochasticNestedProblem(schmidt, b1, T(efficacy = 0.99); area,
 ↪  coverage = 1.0); rng = Xoshiro(s))
    plot!(plt, r.t, r.adults ./ r.adults[1]; color = :steelblue, alpha =
     ↪  0.6, linewidth = 1,
        label = s == 1 ? "stochastic runs" : "")
end
plt
```

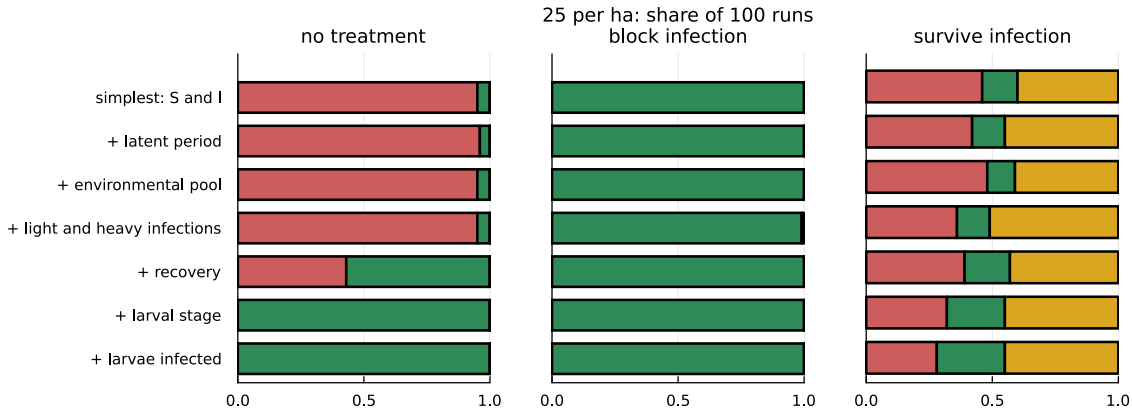


The runs and the equations disagree completely. With 99% of adults protected, the one infected animal infects about a quarter of an animal on average, so Bsal almost always dies out within weeks. In the equations it never quite dies out: a trace of infection, far below one animal, waits until enough untreated young have been born and then sets off a second epidemic. The fade-out rule did not catch this, because its threshold (one animal per 100 km²) is far below one animal in 20 hectares. A population-level effect of the treatment that appeared in every deterministic model was produced by fractions of an animal.

Fates along the ladder

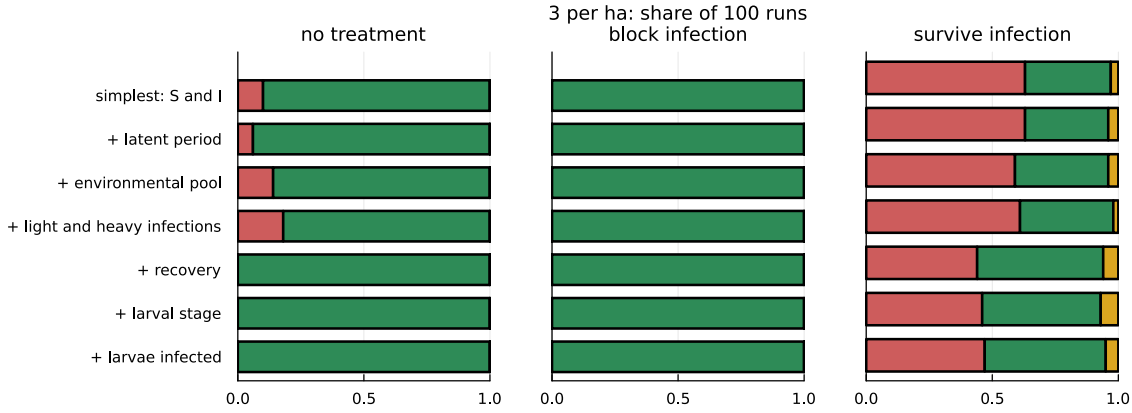
```
function fate_figure(d)
    plots = map(treatments) do (t, _, _)
        rows = [sto[(d, t, i)] for i in 1:length(ladder_bios)]
        k = length(rows)
        groupedbar_data = hcat([r.host for r in rows], [r.gone for r in
↪   rows], [r.both for r in rows])
        bar(1:k, groupedbar_data[:, 1]; orientation = :h, color =
        ↪  :indianred, label = "host lost",
            yticks = (1:k, t == "no treatment" ? names_ : fill("", k)),
↪   yflip = true, xlims = (0, 1),
            xticks = 0:0.5:1, title = t, legend = false, bar_width = 0.7)
        bar!(1:k, groupedbar_data[:, 1] .+ groupedbar_data[:, 2];
        ↪  orientation = :h, color = :seagreen,
            label = "Bsal lost", fillrange = groupedbar_data[:, 1],
↪   bar_width = 0.7)
        bar!(1:k, ones(k); orientation = :h, color = :goldenrod, label =
        ↪  "both persist",
```

```
            fillrange = groupedbar_data[:, 1] .+ groupedbar_data[:, 2],
↪   bar_width = 0.7)
    end
    plot(plots...; layout = (1, 3), size = (900, 330), left_margin =
↪    4Plots.mm,
        plot_title = "$d: share of 100 runs", plot_titlefontsize = 10)
end
fate_figure("25 per ha")
```

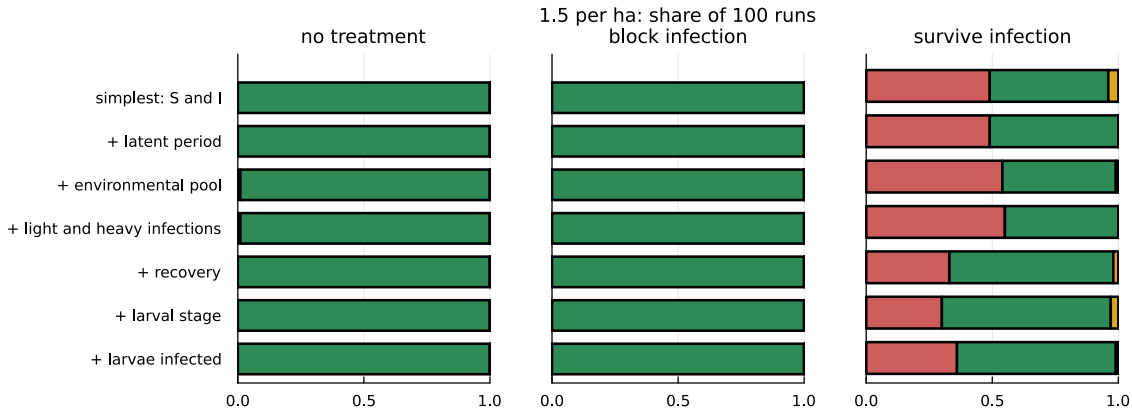


Red: the host population is lost within ten years. Green: Bsal is lost and the host remains. Yellow: both persist.

```
fate_figure("3 per ha")
```




```
fate_figure("1.5 per ha")
```



Mean adults over ten years, as a fraction of the starting number, and the share of introductions that never became an outbreak (fewer than ten animals ever infected at once):

```
println("| Density | Model | None: mean (minor) | Block: mean (minor) |
 ↪  Survive: mean (minor) |")
println("|---|---|---|---|---|")
for (d, _) in densities, (i, n) in enumerate(names_)
    cells = [@sprintf("%.2f (%.2f)", sto[(d, t, i)].mean, sto[(d, t,
 ↪  i)].minor) for (t, _, _) in treatments]
    println("| ", d, " | ", n, " | ", join(cells, " | "), " |")
end
```

Density	Model	None: mean (minor)	Block: mean (minor)	Survive: mean (minor)
25 per ha	simplest: S and I	0.05 (0.05)	1.00 (0.97)	0.17 (0.04)
25 per ha	+ latent period	0.04 (0.04)	1.00 (0.97)	0.17 (0.04)
25 per ha	+ environmental pool	0.05 (0.05)	0.81 (0.63)	0.18 (0.05)
25 per ha	+ light and heavy infections	0.06 (0.05)	0.73 (0.52)	0.18 (0.05)
25 per ha	+ recovery	0.07 (0.05)	0.89 (0.78)	0.18 (0.05)
25 per ha	+ larval stage	0.10 (0.02)	0.90 (0.80)	0.18 (0.05)
25 per ha	+ larvae infected	0.11 (0.04)	0.83 (0.71)	0.19 (0.06)
3 per ha	simplest: S and I	0.50 (0.39)	0.99 (1.00)	0.37 (0.28)
3 per ha	+ latent period	0.47 (0.36)	0.99 (1.00)	0.37 (0.28)
3 per ha	+ environmental pool	0.48 (0.45)	0.99 (1.00)	0.40 (0.31)
3 per ha	+ light and heavy infections	0.39 (0.29)	0.98 (1.00)	0.41 (0.31)
3 per ha	+ recovery	0.76 (0.67)	0.98 (1.00)	0.51 (0.43)
3 per ha	+ larval stage	0.84 (0.70)	0.99 (1.00)	0.39 (0.28)
3 per ha	+ larvae infected	0.81 (0.60)	0.99 (1.00)	0.42 (0.32)

Density	Model	None: mean (minor)	Block: mean (minor)	Survive: mean (minor)
1.5 per ha	simplest: S and I	0.85 (0.91)	0.99 (1.00)	0.51 (0.43)
1.5 per ha	+ latent period	0.83 (0.92)	0.98 (1.00)	0.51 (0.43)
1.5 per ha	+ environmental pool	0.81 (0.98)	0.98 (1.00)	0.49 (0.40)
1.5 per ha	+ light and heavy infections	0.76 (0.93)	0.99 (1.00)	0.51 (0.42)
1.5 per ha	+ recovery	0.96 (1.00)	0.99 (1.00)	0.61 (0.54)
1.5 per ha	+ larval stage	0.97 (1.00)	0.98 (1.00)	0.61 (0.55)
1.5 per ha	+ larvae infected	0.98 (1.00)	0.98 (1.00)	0.58 (0.50)

What chance changes

1. The blocking treatment given on arrival nearly always works at Schmidt's density in the simpler models (about 0.99 of adults kept), against a third in the equations. Pieces of biology that make a chain of infection last longer (an environmental pool, light infections, infected larvae) bring it down to between 0.73 and 0.83, because they give untreated young time to be born before Bsal is gone.
2. Without treatment, the simple models lose the host. At Schmidt's density the host population disappears in almost every run of the four simplest models. With recovery it survives in about half, and with a larval stage in all. The larval refuge seen in the equations is real, not an artefact.
3. Near the threshold most introductions fizzle. At 1.5 per hectare most single introductions never become an outbreak, and with recovery or larvae nearly all of them.
4. The survive-type treatment often loses the host in small populations. In 20 hectares it loses the host in about 30 to 60% of runs at every density. Near the threshold, and in every model with a larval stage, that is more often than with no treatment at all.

Does the size of the population matter?

The same comparison in 200 hectares (ten times as many animals) for three models:

```
println("| Density | Model | Treatment | Host lost: 20 ha | 200 ha | Mean
 ↪  adults: 20 ha | 200 ha |")
println("|---|---|---|---|---|---|---|")
for (d, p) in densities[1:2], i in (1, 5, 7), (t, tr, cov) in treatments
    small, big = sto[(d, t, i)], stochastic_fates(p, ladder_bios[i], tr,
 ↪  cov; area = 200.0)
    @printf("| %s | %s | %s | %.2f | %.2f | %.2f | %.2f |\n", d, names_[i],
     ↪  t, small.host, big.host,
        small.mean, big.mean)
end
```

Density	Model	Treatment	Host lost: 20 ha	200 ha	Mean adults: 20 ha	200 ha
25 per ha	simplest: S and I	no treatment	0.95	0.95	0.05	0.05
25 per ha	simplest: S and I	block infection	0.00	0.00	1.00	0.99
25 per ha	simplest: S and I	survive infection	0.46	0.00	0.17	0.15
25 per ha	+ recovery	no treatment	0.43	0.00	0.07	0.06
25 per ha	+ recovery	block infection	0.00	0.00	0.89	0.86
25 per ha	+ recovery	survive infection	0.39	0.00	0.18	0.18
25 per ha	+ larvae infected	no treatment	0.00	0.00	0.11	0.11
25 per ha	+ larvae infected	block infection	0.00	0.00	0.83	0.88
25 per ha	+ larvae infected	survive infection	0.28	0.00	0.19	0.16
3 per ha	simplest: S and I	no treatment	0.10	0.00	0.50	0.48
3 per ha	simplest: S and I	block infection	0.00	0.00	0.99	1.01
3 per ha	simplest: S and I	survive infection	0.63	0.06	0.37	0.38
3 per ha	+ recovery	no treatment	0.00	0.00	0.76	0.81
3 per ha	+ recovery	block infection	0.00	0.00	0.98	1.00
3 per ha	+ recovery	survive infection	0.44	0.01	0.51	0.42
3 per ha	+ larvae infected	no treatment	0.00	0.00	0.81	0.81
3 per ha	+ larvae infected	block infection	0.00	0.00	0.99	1.00
3 per ha	+ larvae infected	survive infection	0.47	0.01	0.42	0.51

The mean numbers of adults hold. Whether the host is lost does not: in 200 hectares the host is lost only in the simplest untreated model at Schmidt's density, where Bsal kills almost every animal. Under the survive-type treatment, host and Bsal instead persist together. In 20 hectares, the treated animals that survive and keep Bsal going are too few to outlast the untreated young it

kills.

What this shows

1. Whether realism matters depends on the regime. For a pathogen far above its invasion threshold, six pieces of host biology barely change the outcome or the value of treatment. Near the threshold the same pieces change both a great deal.
2. So the first question is where the system sits. Far above the threshold, the deterministic models agree with each other. In sparser populations the answer depends on biology that has not been measured, recovery above all.
3. Effects of single pieces are not transferable. They vary with the rest of the model, so the result of adding one piece to a simple model should not be read as the effect of that piece.
4. Protection that keeps infected animals alive can be worse than none, where the untreated epidemic would have burnt out.
5. A model can mislead through its mathematics as well as its biology. One apparent effect disappeared under a fade-out rule. More importantly, the deterministic models all agreed that blocking infection in every adult on arrival leaves about a third of a dense population, and the stochastic version shows that result was made of fractions of an animal: with whole animals, Bsal nearly always dies out first.
6. Chance is a piece of realism like the others, and here it was the largest. Agreement among the 48 deterministic models at Schmidt's density did not mean the answer was robust: all of them shared the same mathematical assumption.

Limits

- The nested model covers the salamander system only. The toad and frog systems of the first vignette have features it lacks: a seasonal cohort, and a continuous load on each animal.
- The fade-out rule is a threshold on a deterministic model, with an illustrative value. Its threshold was far too low for a population of a few hundred animals; the stochastic version replaces it.
- The stochastic model advances in steps of six hours, approximate for the one-day latent period. Bsal arrives as a single infected adult, and the area (20 hectares) sets the population size; results for treatments that let infected animals survive depend on it.
- Space is not modelled: the area is one well-mixed population.
- Fates are judged within ten years. A population left at a few percent of its former density is counted as remaining.
- Treated animals that survive infection never clear it unless `recovery` is on.
- The values attached to the switches are illustrative. Different values would change the sizes of the effects, though the contrast between regimes should hold.

References

- Schmidt BR, Bozzuto C, Lötters S, Steinfartz S (2017) Dynamics of host populations affected by the emerging fungal pathogen *Batrachochytrium salamandrivorans*. *Royal Society Open Science* 4:160801. doi:10.1098/rsos.160801
- Wang Y, Verbrugghe E, Meuris L, Chiers K, et al. (2021) Epidermal galactose spurs chytrid virulence and predicts amphibian colonization. *Nature Communications* 12:5788. doi:10.1038/s41467-021-26127-9